Akgül et al. 2026 — Rethinking RL for LLM Reasoning

Sparse Policy Selection, Not Capability Learning | arXiv:2605.06241 | my study repo

One-Liner

RL post-training for math reasoning reranks ∼2% of tokens at high-entropy decision points toward alternatives the base model already had in its top-5; an offline contrastive fine-tune gated on base entropy matches full RL at $\sim 1000\times$ less compute.

What's Novel

- Token-level mechanistic claim with a causal probe: oracle correction at reranked positions recovers RL's pass@1 exactly; random controls do not.
- Teacher-free locator: base entropy alone ($\tau \approx 1.4$ nats) identifies the intervention sites without ever consulting the RL model.
- Constructive alternative: *ReasonMaxxer* drops the online RL loop, keeps GRPO's per-group advantage normalisation, applies CE only at decision points, KL-anchors elsewhere.
- RL's parametric delta is rank-32 LoRA on QKVO (0.27–0.49% of params); rank-8 on W_O alone nearly suffices.

Math in 60 Seconds

Token-level predictive entropy under the base model

$$H_t = -\sum_v \pi_{\text{base}}(v \mid q, o_{<t}) \log \pi_{\text{base}}(v \mid q, o_{<t}).$$

Decision-point set: $D = \{t : H_t > \tau\}$. Per-rollout advantage (from GRPO):

$$A_i = \frac{r_i - \bar{r}}{\sigma_r + \epsilon}, \quad r_i \in \{0, 1\}.$$

ReasonMaxxer loss:

$$\mathcal{L} = -\sum_i A_i \sum_{t \in D^{(i)}} \log \pi_\theta\big(o_t^{(i)} \,\big|\, q, o_{<t}^{(i)}\big) \; + \; \lambda \sum_{t \notin D^{(i)}} \mathrm{KL}\big(\pi_{\text{base}} \,\big\|\, \pi_\theta\big).$$

Empirical: reranked fraction 1–4%; the RL-promoted token sits at mean rank ≈ 2.2 under the base; entropy at reranked positions is 5–12× that at unchanged ones.

What I'd Push Back On

- "Within base top-5" is measured on already RL-trained checkpoints; partial circularity until verified on a held-out, truly untouched base.
- Threshold τ is tuned per setup. "No RL needed" partially trades the optimisation loop for a scalar grid-search.
- All wins are on math with verifiable scalar rewards. Open-ended tasks, agentic tool use, or tasks where the right intervention is *low*-entropy fall outside the licensed scope — and the framework predicts its own failure there.
- "1000× cheaper" compares to full RL training only; ReasonMaxxer still needs labeled-correctness rollouts. Accounting could be sharper.

My One Proposed Extension

A Lipschitz bound on rank displacement. The paper observes the RL-promoted token sits at mean rank ≈ 2.2 across all four base/RL pairs — a striking regularity left unexplained. I conjecture that under bounded GRPO learning rate η and per-step KL-to-reference budget β, the expected rank displacement satisfies $\mathbb{E}[\Delta_t \mid t \in R] \leq C\eta/\beta$. The derivation routes through the policy-ratio bound under the standard KL-clipped GRPO update (Shao et al. 2024) and a Lipschitz argument on the softmax-rank-maximiser. Falsifiable with a single training-run sweep across η at fixed β, checking linear scaling. Closes the loop between the empirical mean-rank observation and the underlying optimisation dynamics, and would extend cleanly to predict where the top-5 assumption breaks under higher-learning-rate regimes.

Full proposal set: `05-improvements.tex` in the study repo.

My One Question

If the entire RL behavioural delta is captured by a rank-32 LoRA on W_O, why train at all? What's the obstacle to replacing the training stage with a search over inference-time steering vectors applied only at high-entropy positions?